

Rajjat Dadwal, PhD

MACHINE LEARNING · MULTIMODAL & LLM RESEARCH · GEOSPATIAL AI

Bonn, Germany · Germany Permanent Residence (Niederlassungserlaubnis)

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Profile

Senior Research Scientist with a PhD in Artificial Intelligence and 8 years of experience building machine learning systems for large-scale, real-world applications. Research spans multimodal learning, graph neural networks, reinforcement learning, and representation learning, with a strong track record of translating research into production-oriented systems: an evaluation framework for LLMs on spatial reasoning and geospatial question answering, a route recommendation model exported to ONNX and deployed on an industry partner's infrastructure, and a multimodal architecture that improved downstream prediction accuracy by 19%. Experienced collaborating with academic and industry partners (Volkswagen, EU-funded projects) and mentoring PhD, Master's, and Bachelor's students.

Core Competencies

- **Machine Learning & Deep Learning:** Python, PyTorch, TensorFlow, multimodal learning, graph neural networks, reinforcement learning, representation learning
- **LLM Evaluation & Fine-Tuning:** Evaluation frameworks surfacing model failure modes; fine-tuned LLMs for applied QA tasks; basic DeepSpeed for distributed training
- **Recommender Systems:** Personalized route recommendation (RL-based); point-of-interest recommendation from historical trajectory data
- **Production ML:** Exporting models to ONNX; supporting industry-partner deployment; scalable pipelines from research prototype to deployment-ready systems
- **Geospatial & Data Engineering:** QGIS, GeoPandas, PostGIS, Apache Spark, Hadoop (HDFS), Kafka, distributed RDF processing (SANS Stack)

Professional Experience

DSIS, University of Bonn

Bonn, Germany

SENIOR RESEARCH SCIENTIST & PROJECT LEAD

2023-Present

- Designed and built an evaluation framework for large language models on spatial reasoning and geospatial question-answering, surfacing model failure modes; fine-tuned LLMs for the task, with basic working experience using DeepSpeed for distributed training
- Leading the MoToRes AI research team, developing a reinforcement-learning-based personalized route recommendation model; exported the model to ONNX format and supported an industry partner in deploying it on their own server infrastructure
- Developed a point-of-interest (POI) recommendation model predicting places of interest for individual users based on historical GPS trajectory data (approximately one year of work)
- Developed a multimodal deep learning architecture (PyTorch) fusing heterogeneous data sources into adaptive region-level embeddings, improving downstream prediction accuracy by 19%; published as first-author research at PAKDD 2024
- Built scalable ML pipelines integrating structured, unstructured, and knowledge-based data for end-to-end predictive systems; collaborating with academic and industry partners to translate research outcomes into deployable prototypes
- Supervising and mentoring PhD, Master's, and Bachelor's students in data analytics and machine learning projects

L3S Research Center

Hannover, Germany

RESEARCH SCIENTIST

2019-2023

- Led the CampaNeo project, an industrial-academic AI collaboration with Volkswagen on large-scale predictive analytics for intelligent mobility
- Developed an RNN-based model for spatiotemporal accident prediction, improving performance by 4 percentage points through systematic benchmarking of ensemble and deep learning approaches
- Designed a privacy-preserving GPS trajectory watermarking framework (EU project SmashHit) achieving 99% recognition accuracy while preserving data utility

Fraunhofer IAIS

Sankt Augustin, Germany

DATA SCIENTIST

2018-2019

- Developed scalable distributed processing pipelines for large-scale geospatial and graph-based datasets (TomTom) using Spark and Scala
- Developed anomaly detection methods for knowledge graph analytics using the SANS Stack and distributed RDF processing

Reliance Jio Infocomm Ltd.

Mumbai, India

SOFTWARE DEVELOPER

2013-2015

- Built scalable web services for real-time and batch analytics pipelines, handling high-volume mobile network data (approximately 100,000 users)
- Developed an element management system for managing fault, configuration, and performance of 4G network entities in C/C++

Education

University of Bonn

Bonn, Germany

PHD IN COMPUTER SCIENCE / ARTIFICIAL INTELLIGENCE

2019-2025

- Thesis: "Adaptive Geospatial Data Representation for Data Analytics and Sharing in the Mobility Domain." Research spanning multi-modal learning, representation learning, and graph neural networks. Magna cum laude (1.1)

University of Bonn

Bonn, Germany

M.SC. IN COMPUTER SCIENCE

2015-2018

- Thesis: "Scalable Numerical Outlier Detection in Knowledge Graphs," published at EKAU 2018. Grade: 1.9

National Institute of Technology, Hamirpur

Hamirpur, India

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE & ENGINEERING

2009-2013

- CGPA: 7.85/10

Selected Publications

- Rajjat Dadwal, Damien Graux, Gezim Sejdiu, Hajira Jabeen, Jens Lehmann (2019). Clustering pipelines of large RDF POI data. ESWC.
- Rajjat Dadwal, Thorben Funke, Elena Demidova (2021). An adaptive clustering approach for accident prediction. ITSC / IEEE Access.
- Rajjat Dadwal, Thorben Funke, Michael Nüsken, Elena Demidova (2022). W-Trace: Robust and Effective Watermarking for GPS Trajectories. ACM SIGSPATIAL.
- Rajjat Dadwal, Ran Yu, Elena Demidova (2024). A multimodal and multitask approach for adaptive geospatial region embeddings. PAKDD.
- Rajjat Dadwal, Thorben Funke, Michael Nüsken, Elena Demidova (2025). Towards Effective, Robust and Utility-preserving Watermarking of GPS Trajectories. TSAS Journal.
- Amitoz Azad, Rajjat Dadwal, Elena Demidova (2026). Learning Generalized Hessian on Graphs for Continuous Graph Learning. ECML PKDD.
- Genivika Mann, Rajjat Dadwal, Elena Demidova (2026). DIRECT: Deep Reinforcement Learning for Tourist Route Generation. ACM TSAS Journal and SigSpatial conference (To appear).
- Joshua Petteño, Rajjat Dadwal, Alishiba Dsouza, Elena Demidova (2026). MOSAIC: Synthetic Trajectory Generation with Reinforcement Learning. ACM SIGSPATIAL (Poster Paper, accepted).

Academic Service

2026-2027 **Program Committee Member**, ISWC, ECML PKDD, AAAI

2019-2025 **Reviewer**, NeurIPS, ICML, ISWC, ESWC, ACM SIGSPATIAL, WWW

Languages

- English: Fluent German: B1 (certified) Hindi: Native

Honors and Awards

2009 **Top 0.07% (Rank 78/100,000)**, National University Entrance Exam, National Institute of Technology, Hamirpur

2023 **Selected for a research visit (one of three students)**, University of Melbourne, Australia